

EXP 30 – Backward Chaining (Prolog)

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Aim

To write a Prolog program that demonstrates backward chaining by checking whether an object is a bird or a flying bird based on predefined facts and rules.

Algorithm

1. Define basic facts such as has_feathers, lays_eggs, and can_fly.
2. To determine if it is a bird:
 - Check if it has feathers.
 - Check if it lays eggs.
 - If both are true, conclude it is a bird.
3. To determine if it is a flying bird:
 - First confirm it is a bird.
 - Then check if it can fly.
4. Use backward chaining: Prolog tries to satisfy goals by checking rules and facts backward from the query.

Source Code

```
prolog
```

```
fact(has_feathers).
```

```
fact(lays_eggs).
```

```
fact(can_fly).
```

```
is_bird :-
```

```
    format("Checking if it is a bird...~n"),
```

```
    fact(has_feathers),
```

```
    fact(lays_eggs),
```

```
format("Confirmed: it is a bird~n").
```

is_flying_bird :-

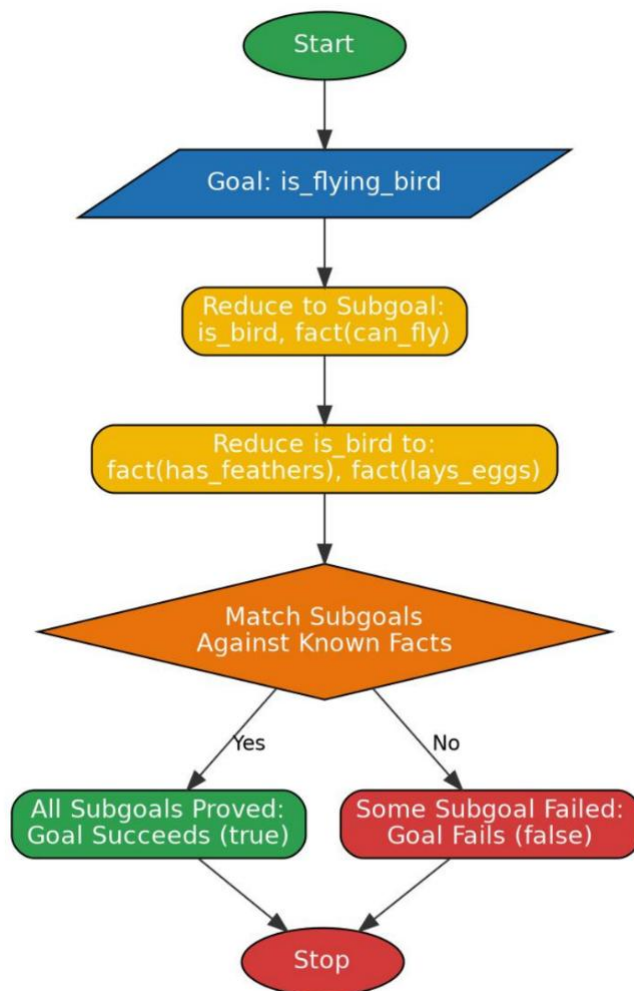
```
format("Checking if it is a flying bird...~n"),
```

```
is_bird,
```

```
fact(can_fly),
```

```
format("Confirmed: it is a flying bird~n").
```

Flowchart:



Output:

```
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?- cd('C:/Users/yanar/Documents/CSA1717').
true.

?- [backward].
true.

?- is_bird.
Checking if it is a bird...
Confirmed: it is a bird
true.

?- is_flying_bird.
Checking if it is a flying bird...
Checking if it is a bird...
Confirmed: it is a bird
Confirmed: it is a flying bird
true.
```

Result

The program successfully verifies whether the object is a bird and further checks if it is a flying bird using backward chaining, confirming both conclusions based on the available facts.